



RAG assistant for Slovenian employment law

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Abstract

We built a question-answering assistant for Slovenian employment law using retrieval-augmented generation. We started from the COLESLAW legal corpus, ran into three systematic failures caused by outdated and low-authority documents, and fixed them by rebuilding the knowledge base from current official sources and iterating through five system prompt versions. The final system retrieves from PISRS statutes, government guidance, and related agencies, ranks sources by authority, and uses a strict Slovenian prompt that requires citations and refuses out-of-scope questions. On a 40-question evaluation set it achieves perfect retrieval with no false evidence for unanswerable questions, and the full RAG setup scores 2.95/5 for answer correctness and 1.00 for both citation support and refusal accuracy, compared to 0.50 and 0.00 for the baseline without any retrieved context. We also prepared fine-tuning data but ultimately did not use it, since for a legal assistant current official sources matter more than what the model has memorised. The deployed assistant is available at <https://ai.koderverse.com/>.

Keywords

Slovenian employment law, legal question answering, retrieval-augmented generation, RAG, source grounding

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Motivation

Language models are not reliable for legal questions. They often sound confident while citing articles that do not exist, mixing up rules from different years, or applying foreign law to a Slovenian question. This is a real problem: employment law is one of the areas people most often want quick guidance on, and the questions have correct answers that can be verified against published statutes. We picked Slovenian employment law as our domain because it is well scoped, the official sources are publicly available online, and there are already a few Slovenian legal chatbots showing that people want this kind of tool [1, 2].

We designed the assistant to be informational rather than a legal advisor. It only answers questions about Slovenian employment relationships—contracts, termination, leave, wages, sick leave, and related topics—and it refuses anything else. It also refuses when the retrieved documents do not contain enough information to answer the specific question. This strict scope was deliberate: it makes evaluation meaningful because we can test not just whether the system answers correctly, but also whether it knows when not to answer.

Related Work

The core idea we used is retrieval-augmented generation: instead of relying on what the model memorised during training, you retrieve relevant documents at query time and give them to the model as context [3]. For a legal assistant this is ideal because a claim grounded in a retrieved article can be checked against the source, while a fluent answer from model memory alone cannot.

We also considered fine-tuning as an alternative. Techniques like LoRA can teach a model new output formats or styles without retraining the whole network [4], and tool-using approaches let models call external systems to get fresh data [5]. The trade-off between all these methods—prompting, RAG, and fine-tuning—has been surveyed recently in the context of injecting domain knowledge [6]. We went with RAG because legal answers need to be grounded in current statutes, and that requires the actual text to be present at generation time rather than baked into model weights.

For the legal text corpus we started with COLESLAW 1.0, a large collection of Slovenian legal documents including statutes, court decisions, and administrative texts [7].

Corpus and Sources

Our first approach was to filter the COLESLAW corpus down to employment-law relevant text using keyword matching and use that as the knowledge base. It gave short retrieved contexts (around 190 words per question on average) and we quickly found three concrete answer-quality failures: one question about fixed-term contract duration surfaced the wrong ZDR-1 article, another about who pays sick leave after day 30 returned outdated wording from an older version of the rule, and a third ambiguous question about leave entitlement produced a confident general answer instead of flagging the missing details. The underlying problem was that COLESLAW is full of court decisions and older documents: there is plenty of text mentioning employment topics, but the current statutory text is not always the top result, and short context windows leave no room for corroborating sources.

To fix this we built a new knowledge base directly from the official online sources. We fetched and chunked documents from PISRS (the official Slovenian legal register), GOV.SI and MDDSZ (the main government and labour ministry guidance pages), ZZZS (health insurance), IRSD, eUprava, SPOT, ESS, and OPSI. Each chunk was tagged with a source type so the retriever could prefer higher-authority sources. We also kept a small set of COLESLAW court decisions as low-priority support for questions that specifically ask about legal interpretation or past court practice. The final corpus has 1,371 chunks in total: 1,059 primary-law chunks from PISRS statutes, 189 official interpretation chunks, 93 operational-guidance chunks, and 30 case-law chunks.

The authority order we enforced at both retrieval and prompt level is:

1. Statutes from PISRS, especially ZDR-1 (the Employment Relations Act) [8].
2. Official guidance from GOV.SI and MDDSZ [9, 10].
3. Operational guidance from ZZZS, eUprava, SPOT, ESS, and OPSI [11, 12].
4. Court decisions from sodnapraks.si and COLESLAW, used only as supporting context.

Moving to this official corpus fixed all three earlier failures and grew average context length from around 190 to 1,430 words per question. We also built a source monitor that checks whether all 37 official sources are still reachable, so the corpus can be kept up to date as laws change.

System

For retrieval we use BM25, a standard lexical ranking algorithm, with a simple term-overlap fallback if the BM25 library is not available. We apply Slovenian-specific preprocessing: common stopwords are removed and a light rule-based stemmer handles the most frequent inflection patterns in the language.

On top of the raw BM25 score we apply two domain-specific adjustments. The first gives a score bonus to higher-authority source types: primary law gets the biggest boost, then official interpretation, then operational guidance. Case-law chunks receive a large negative penalty unless the question explicitly asks about court practice or legal interpretation, so that a question about the letter of the law does not accidentally surface a court case as the top result. The second adjustment handles actor-sensitive questions. Employment law questions are often asymmetric—what an employer can do and what an employee can do are different things—so we add small boosts when the retrieved text specifically matches the actor mentioned in the question.

The system prompt took five iterations to get right. The first version established the basic rule: answer only from the retrieved context and refuse if you cannot. The second added an explicit check step so the model thinks about whether context actually supports the answer before generating it. The third focused on handling vague questions, since some questions like “How much leave do I get?” cannot be answered without knowing more details. The fourth added source priority to the prompt instructions, mirroring what the retrieval already does. The fifth and final version added three things: a requirement that every non-refusal answer must cite a specific article or source, a rule that vague questions must start with an explanation of what information is missing rather than a confident general answer, and a rule that when the law has multiple tiers or thresholds the answer must list all of them rather than just mentioning the first.

For the model itself we ran a side-by-side arena evaluation comparing llama3, Mistral 7B, Gemma 3 4B, Qwen 2.5 Coder 7B, and a larger Qwen model. On the short smoke test with RAG context every model scored 5.00/5 for correctness and grounding; without context they all dropped to around 4.2–4.4/5. The main takeaway from this comparison was that retrieval quality and the system prompt mattered much more than which specific model we used.

On the recommendation of the course assistant we also evaluated GaMS-2B-Instruct, a Slovenian-focused model from the GaMS family [13]. GaMS-2B-Instruct tells a notably different story from the general-purpose models. Its baseline correctness without any retrieved context (1.65/5) is more than three times higher than the llama3 baseline (0.50/5), because its Slovenian-specific pretraining already encodes a meaningful amount of employment-law knowledge. With RAG context, correctness rises only modestly to 1.93/5, and refusal accuracy actually drops to zero—the model answers every question even when it should decline. This reveals a clear trade-off: Slovenian-language pretraining gives GaMS a head start on raw knowledge, but the model does not follow strict prompt guardrails as reliably as the general-purpose models do. Citation discipline is also poor, with only 6% of RAG answers including a verifiable source reference compared to 100% for Mistral. For the deployed system we use Mistral 7B because it handles the structured prompt instructions well and

achieves the best overall RAG performance. The deployed assistant is at <https://ai.koderverse.com/>, using deterministic settings (temperature 0) so answers are reproducible.

We also prepared a fine-tuning dataset—176 training examples and 44 validation examples in a chat format—to explore whether LoRA adaptation would help. After looking at the results, we decided not to use it for the final system. The main reasons were that our evaluation showed retrieval quality was the bottleneck rather than output style, that a fine-tuned model’s knowledge of the law would go stale as statutes change, and that running 7B model fine-tuning locally was not practical with the available hardware.

Evaluation

We kept retrieval and answer quality as separate measurements, which turned out to be important. A retrieval failure (the wrong article is returned) and a generation failure (the model ignores a correct article) produce the same wrong answer but need different fixes. By measuring retrieval with its own metrics first, we could tell whether a bad answer was caused by a corpus gap or by the model not using the context correctly.

For retrieval we measured two things. *Hit@3* checks whether the chunk containing the correct answer is among the top three results the retriever returns. The false-evidence rate checks the opposite failure mode: for questions that have no answer in the corpus, does the retriever still return a chunk that looks relevant? A high false-evidence rate would trick the model into generating a confident but unsupported answer for a question it should refuse.

For answer quality we used seven metrics. *Correctness* (0–5) measures how accurately the answer matches the expected legal answer—0 means completely wrong or invented, 5 means fully correct with all relevant details. *Grounding* (0–5) measures how well the answer stays within the retrieved documents rather than drawing on model memory; a well-grounded answer would not change if the model’s training data changed. *Completeness* (0–5) measures how fully the answer covers the question, including thresholds, exceptions, and edge cases mentioned in the law. *Clarity* (0–5) measures how readable and well-structured the answer is, independent of whether it is correct. *Hallucination* (0–5) measures how much the model invents information not supported by the retrieved context—here 0 is best and 5 is worst, so this scale runs in the opposite direction from the others. *Citation support rate* (0–1) is the fraction of answers that include at least one verifiable reference to a specific article or official source. *Refusal accuracy* (0–1) is the fraction of deliberately unanswerable questions that the model correctly declines to answer rather than guessing.

The evaluation set has 40 questions covering twelve topic areas: annual leave and regres, fixed-term contracts, termination, severance, sick leave, minimum wage, working-time records, occupational safety, unpaid wages, collective agreements, court interpretation questions, and questions that are

deliberately ambiguous or out of scope. That last category is important: an assistant that answers confidently even when it should not know is dangerous for a legal domain, so we treated refusal accuracy as a first-class metric.

Table 1. Retrieval results. Higher Hit@3 and lower false evidence are better. The official corpus was evaluated on the full 40-question set; the initial COLESLAW slice was measured on a 5-question smoke test covering only answerable questions. The main practical difference is context length.

Corpus	Hit@3	False ev.	Ctx. words
Official corpus	1.000	0.000	1430.8
Initial COLESLAW slice	1.000	0.000	189.9

Table 2. Answer quality on the 40-question set. Baseline rows have no retrieved context. Strict RAG uses the official corpus, the final system prompt, Mistral 7B, and temperature 0. GaMS-2B is the Slovenian GaMS-2B-Instruct model with the same corpus and prompt. Correctness and hallucination are 0–5; citations and refusal are fractions.

Variant	Correct.	Halluc.	Citations	Refusal
Baseline (llama3)	0.50	4.12	0.00	0.00
Strict RAG (Mistral 7B)	2.95	1.68	1.00	1.00
GaMS-2B baseline	1.65	3.60	0.00	0.20
GaMS-2B RAG	1.93	2.75	0.06	0.00

Table 3. The three evaluation questions that failed with the initial COLESLAW corpus and how they were fixed.

Question	Fix
q009: fixed-term contract duration	ZDR-1 art. 55 retrieved first; two-year limit cited
q012: sick-pay payer after day 30	Current 30-working-day / 31st-day rule retrieved from both ZDR-1 and ZZZS
q014: ambiguous leave question	Ambiguity warning issued; ZDR-1 arts. 159 and 161 cited

For Mistral 7B the improvements from adding RAG are substantial across every metric. Grounding rises from 0.12 to 4.83, meaning the model shifts from answering almost entirely from memory to drawing almost entirely from the retrieved text. Citation support goes from 0.00 to 1.00, and refusal accuracy from 0.00 to 1.00: without context the model freely invents legal provisions and never refuses; with context and the strict prompt it cites sources and correctly turns away every question it cannot answer. Answer correctness rises from 0.50 to 2.95 out of 5, though a gap remains between how well the right sources are retrieved and how well the model synthesises them into a complete answer. Finding the right article is one step; combining information across several statutory paragraphs into a concise correct answer is harder.

The GaMS-2B rows show a different pattern. Without context, GaMS-2B scores 1.65 for correctness—more than

three times the llama3 baseline—because its Slovenian pre-training already encodes some employment-law knowledge. With RAG context, correctness rises only slightly to 1.93, and refusal accuracy drops to zero: the model answers every question even when it should decline. This suggests that Slovenian-language pretraining helps with raw language quality but does not substitute for instruction-following discipline. Fine-tuning GaMS on the strict RAG prompt format would be the natural next step to combine its language strength with the refusal and citation behaviour that the final system achieves with Mistral.

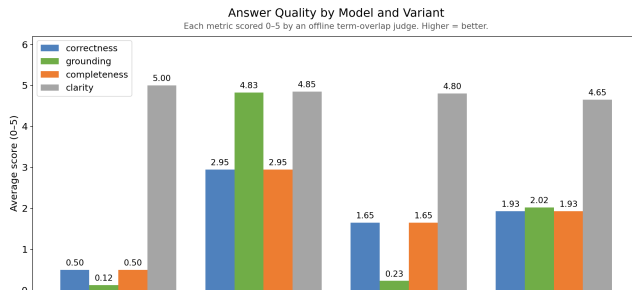


Figure 1. Correctness, grounding, completeness, and clarity (all 0–5, higher is better) for the four evaluated variants. Strict RAG (Mistral 7B) shows large gains on correctness and grounding over the baseline. GaMS-2B baseline already scores higher on correctness than the llama3 baseline, reflecting Slovenian-language pretraining, but gains less from retrieved context than Mistral does.

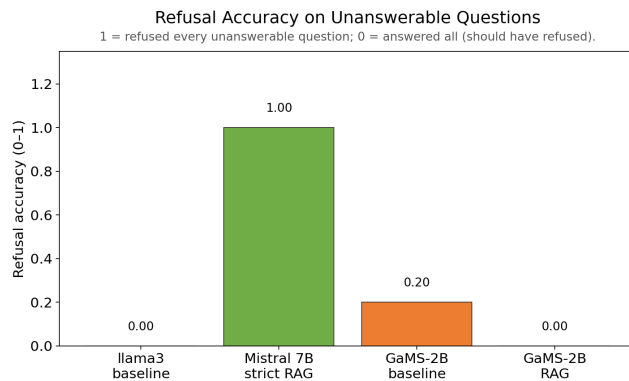


Figure 2. Refusal accuracy (fraction of unanswerable questions correctly declined) for all four variants. A score of 0.00 means the model answered every question it should have refused; 1.00 means it correctly declined all of them. Strict RAG with Mistral is the only variant that achieves perfect refusal. GaMS-2B baseline partially refuses (0.20) but drops to 0.00 when given retrieved context, a key weakness.

Reproducibility and Compliance

Everything needed to reproduce the reported numbers is in the repository. Running one make command rebuilds the

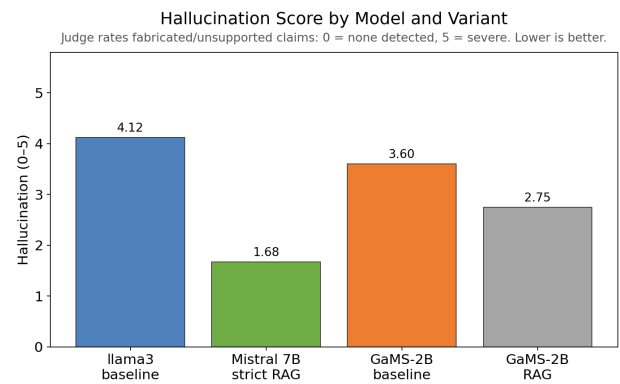


Figure 3. Hallucination score (0–5, lower is better) for all four evaluated variants. A score of 0 means no invented content; 5 means the answer is mostly fabricated. Retrieved context reduces hallucination substantially for both models: Mistral drops from 4.12 (baseline) to 1.68 (RAG), and GaMS-2B drops from 3.60 to 2.75. Mistral strict RAG achieves the lowest hallucination overall.

corpus from official sources, reruns the retrieval and answer evaluations, regenerates charts, and compiles the report PDF. The corpus chunks, evaluation questions, and result snapshots are committed so reviewers can inspect the numbers without rerunning anything. Configuration examples with placeholder values are included; secrets and endpoint addresses are never committed. The raw COLESLAW archive is not in the repository—it is too large—but the scripts for filtering and converting it are included along with the derived chunks we actually used.

The project covers the required course criteria: a strict use case definition, curated official-source knowledge base, one implemented solution (RAG), automated evaluation of both retrieval and answer quality, a final report, and a reproducible repository. Fine-tuning is included as an exploratory artifact that documents why we did not end up using it.

Limitations and Future Work

The biggest gap we did not close is expert legal validation. The evaluation catches retrieval failures and measures citation discipline, but it does not tell us whether the answers are actually correct under Slovenian law. The 2.95/5 correctness score gives a rough sense of how often the answers are in the right direction, but that is not the same as a lawyer reviewing the outputs. The system is informational only.

On the technical side, the retriever uses lexical matching, which means it can miss a question that asks about the same rule in different words. Dense retrieval or a learned reranker trained on employment-law question-document pairs would be the next improvement to try. We held off because it would require a GPU for training and the lexical retriever already achieves perfect Hit@3 on our evaluation set, so the gains would only show up on harder paraphrase cases we have not tested yet.

Fine-tuning GaMS-2B on the strict RAG prompt format would be the most direct next step. Teaching it to refuse unanswerable questions and cite sources would combine its Slovenian language quality with the instruction-following behaviour the final system achieves with Mistral.

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